

ML Assignment 1 - Polynomial Regression

IMT2024031 | Final L1/L2 Regularized Models

1. Final result

The two regression problems were evaluated using polynomial features with regularization. L1 (Lasso) and L2 (Ridge) were compared during cross-validation, and the final configurations were selected from the lowest-validation-error candidates. The final selected models are Lasso for both problems.

Problem	Model	Degree	Alpha	OOF MSE	OOF R2	Non-zero
var1	Lasso (L1)	5	0.00788046	0.327877	0.965743	149 / 461
var2	Lasso (L1)	11	0.00045000	0.235043	0.994675	163 / 363

The reported performance metrics are based on out-of-fold predictions, so they measure predictions on samples that were not used to fit the corresponding fold model. Training-set metrics are included only as a diagnostic, not as the primary performance claim.

2. Methodology

- **Polynomial expansion.** Each original feature set was expanded to all monomials up to the selected degree, without adding a separate constant feature.
- **Standardization.** Polynomial columns were standardized using training-fold statistics before regularized fitting.
- **Regularization comparison.** L1 regularization uses Lasso; L2 regularization uses Ridge. Regularization strength (alpha) was selected by five-fold shuffled cross-validation.
- **Final fitting.** After model selection, the selected Lasso configuration was refit on all 1000 training rows and used to generate the 1000 test predictions.

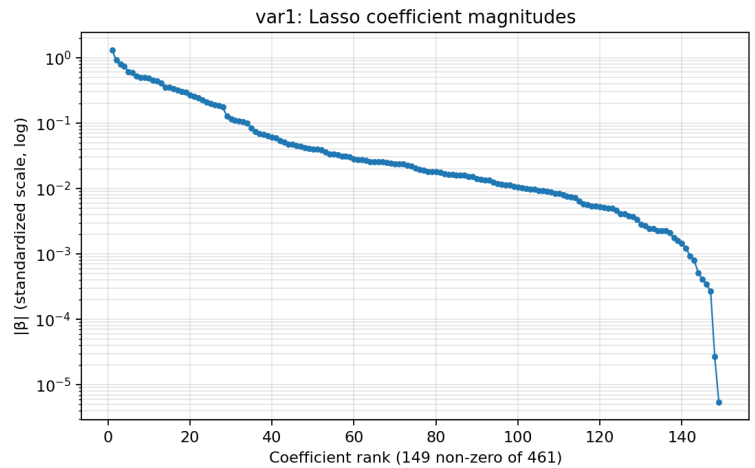
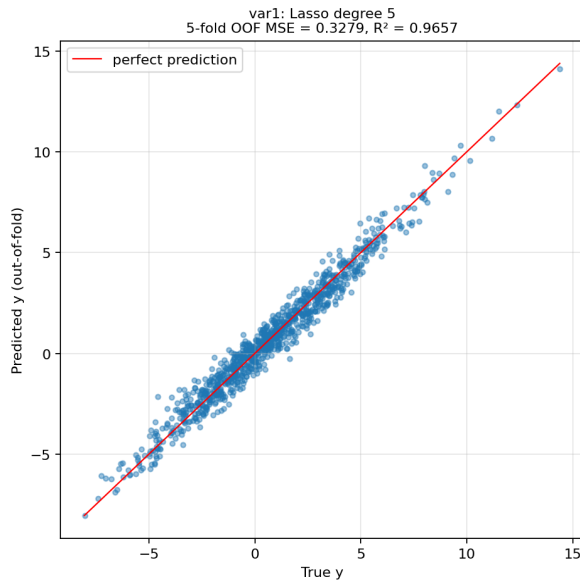
3. Dataset and implementation

Both problems contain 1000 training rows and 1000 test rows. var1 has 6 input features and var2 has 3 input features. The final implementation uses Python, NumPy, pandas, scikit-learn, and matplotlib. The submission package contains the data, final training script, model-selection script, diagnostic plotting code, prediction files, and this report.

GitHub: https://github.com/Heer-create-igtm/ml_assignment_1

4. var1 results

The selected var1 model is a degree-5 Lasso polynomial regression. It expands the six original features into 461 polynomial terms, of which 149 have non-zero coefficients after L1 regularization.



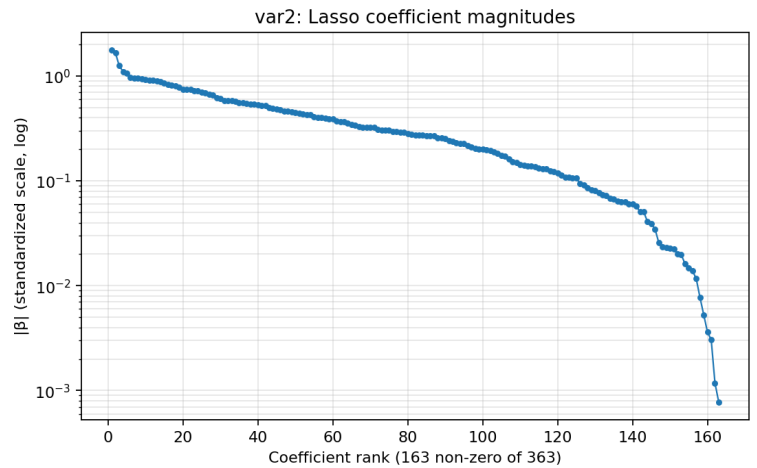
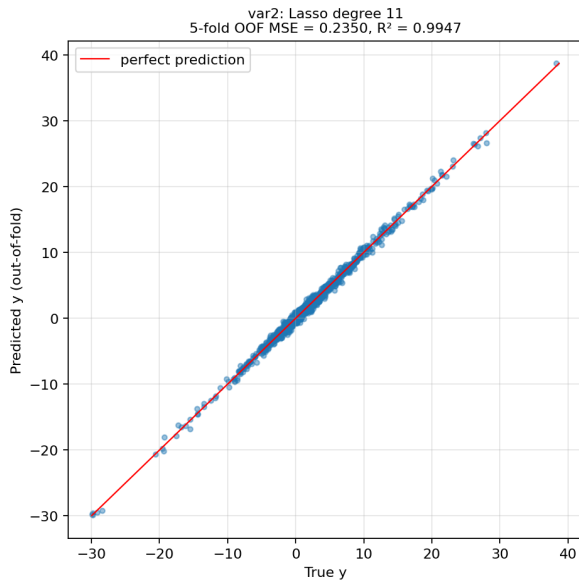
5-fold OOF MSE: 0.327877 **5-fold OOF R2:** 0.965743 **Training MSE:** 0.232103 **Training R2:** 0.975750

The predicted-versus-actual plot shows the OOF predictions close to the ideal diagonal. The coefficient-size plot shows the sparsity produced by the L1 penalty, with many polynomial terms driven to exactly zero.

The final test prediction file is **IMT2024031_pred_var1.csv** and contains exactly 1000 rows with one column, **y**.

5. var2 results

The selected var2 model is a degree-11 Lasso polynomial regression. The three original features expand to 363 polynomial terms, with 163 non-zero coefficients retained after L1 regularization.



5-fold OOF MSE: 0.235043 **5-fold OOF R2:** 0.994675 **Training MSE:** 0.163966 **Training R2:** 0.996285

The OOF scatter is tightly concentrated around the perfect-prediction diagonal, consistent with the high R2. The coefficient-size plot again shows the sparsifying effect of L1 regularization.

The final test prediction file is **IMT2024031_pred_var2.csv** and contains exactly 1000 rows with one column, **y**.

6. Verification and reproducibility

Check	Result
var1 prediction rows	1000; column = y
var2 prediction rows	1000; column = y
var1 prediction reproduction	max abs difference 2.1e-14
var2 prediction reproduction	max abs difference 1.2e-13
Final model implementation	PolynomialFeatures -> StandardScaler -> Lasso
Input data	IMT2024031 train/test CSVs included in package

7. Submission package contents

The final ZIP contains the two prediction CSVs, the final model summary, out-of-fold prediction tables, diagnostic figures, the final training script, the L1/L2 model-selection script, plotting code, requirements, the four input datasets, README instructions, and this report.

The prediction CSVs supplied for submission were directly checked against the final training pipeline. The differences are at floating-point roundoff level only, confirming that the packaged predictions correspond to the stated models and hyperparameters.

Final selected configurations: var1 = Lasso, degree 5, alpha 0.00788046; var2 = Lasso, degree 11, alpha 0.00045.